

REEFPRINT / KHANYA: reading the ore before the mill.

Belt hyperspectral hardness, microscope mineralogy and human-approved plant decisions for South African concentrators

Technical, business and pilot report · Mintek–SCi Grad Hackathon 2026 · Problem 3: Computer vision for real-time mineralogical characterisation

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Abstract

South Africa mined about 71% of the world’s platinum in 2025, yet its concentrators still set mill feed, grind and reagents before the laboratory can say what was in the ore. REEFPRINT/KHANYA reads the ore at three speeds and turns what it reads into one decision that a person approves. On the belt, a hyperspectral camera feeds a model that predicts the ore’s grinding hardness (Bond work index) with a safety upper bound, in 92 ms per parcel on an ordinary processor. On 146 public drill-core samples the model explained 48% of the variation in hardness, against 21% for a colour camera, and a mill feed-rate policy built on its bound raised simulated throughput by 1.9% (95% CI 0.9–2.9%) without more overloads. In the laboratory, a microscope model identifies pyrrhotite, chalcopyrite and pentlandite (IoU 0.87, 0.58 and 0.55 on held-out sections) and gave no unsafe advice on 12 sections. All results come from public data and a simulated plant. At the output of Northam’s Zondereinde mine, each 1% more recovered metal is worth R109–153 million a year in revenue, so a one-belt pilot costing R8–49 million installed breaks even at 0.02–0.21% more recovered metal. We ask the Technology Innovation Agency, Mintek and one producer for a nine-month pilot: one belt, one truth laboratory, with fees paid only on measured value.

Keywords: hyperspectral imaging; Bond work index; process mineralogy; PGM concentrators; pilot business case; advisory process control

1 Introduction

As graduate engineers working on Mintek’s Problem 3, our responsibility is to give a concentrator’s shift team a reliable reading of its ore before that ore reaches the mill, and to turn the reading into a plant setting that a person can approve. This report sets out what we built, how we tested it, what it would cost and earn, and how a funded pilot would prove it on South African ore. It is written for metallurgists, mineralogists, process engineers and the people who fund their technology, so the technical detail is limited to what a decision needs.

1.1 Problem statement

The brief states the problem plainly: “without real-time feedback, processing plants cannot adjust to changes in ore quality, leading to inefficient chemical usage and lower mineral yields” [1]. Ore reaches the mill within minutes of leaving the belt, whereas assays and automated mineralogy are batch laboratory work. Mintek’s own Minerals Processing and Characterisation Cluster runs QEMSCAN [2], and a commercial laboratory describes mineralogy as “not an overnight assay service” [3]. Consequently, each shift sets the mill and the flotation circuit for ore whose mineralogy it has not yet seen.

The cost of deciding blind falls most heavily on South Africa. Figure 1 illustrates world PGM mine production in 2025, highlighting that South Africa mined an estimated 120 of the world’s 170 tonnes of platinum (about 71%) and 70 of 190 tonnes of palladium. South Africa also holds 63,000

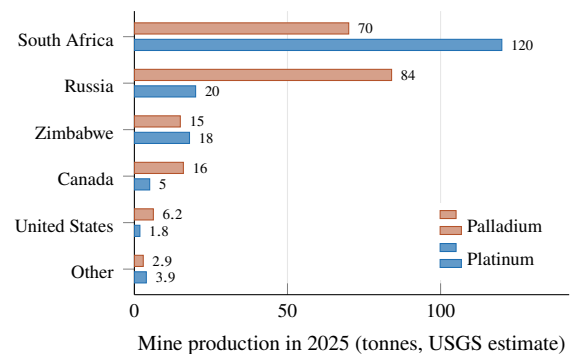


Figure 1: World platinum and palladium mine production in 2025. South Africa mined 120 of 170 tonnes of platinum; Zimbabwe’s Great Dyke ores are of the same family. Data: USGS Mineral Commodity Summaries 2026 [4].

of the world’s more than 76,000 tonnes of PGM reserves, mostly in the Bushveld Complex [4]. The same source attributes the 9% fall in South African PGM output in 2025 to lower palladium prices, higher deep-level mining costs and ongoing disruptions to the electricity supply. Zimbabwe, the third-largest platinum producer, mines the same family of ores on the Great Dyke.

Recovery is where the money is. UG2 ore carries about 30% Cr_2O_3 against about 0.1% for Merensky ore, and the smelter caps the chromite in concentrate. Mintek testwork made UG2 concentrate at 87% recovery with 2.9% Cr_2O_3 , and recoveries above 90% were possible only if the Cr_2O_3 limit was relaxed [5]. Producers report recovery moves of this size to investors: Valterra raised concentrator recoveries

Table 1: The brief’s requirements and the judging criteria, and where this report evidences each.

Requirement	Evidence (section)
<i>The brief</i>	
≥ 3 mineral phases	Pyrrhotite, chalcopyrite, pentlandite (4.4)
Accuracy report	Belt: R^2 , interval coverage, baselines. Microscope: per-class, per-section and decision-level results (4.1–4.5)
Adjust a plant parameter	Mill feed rate from the hardness bound; regrind advice over OPC UA to a simulator (2.3, 4.2)
Real time	92 ms per belt parcel; 17–21 s per microscope scan (3)
<i>Judging criteria</i>	
Innovation	A bound-based feed-forward input that refuses when unsure, with human approval and a post-quantum-signed record (2, 3)
Feasibility	Pricing, support, maintenance, compliance, training and risks (6, 7)
Impact	R109–153 million a year per 1% more recovered metal; break-even 0.02–0.21% (6.3)
Technical execution	Out-of-fold tests, pass marks set before each run, physics checks (4); 515 automated software tests (3)
Presentation clarity	Plain language; a figure or table for each step; every assumption labelled

by 1.2–1.7 percentage points at Mototolo and Amandelbult in 2025 [6]. Comminution, which sets both liberation and mill throughput, used 52% of site electricity in a benchmark of mines [7], on a grid that is itself a constraint.

Thus, the problem is to read the ore before the mill, state how uncertain the reading is, and turn it into a safe, approved setting at a cost a producer can recover.

1.2 Our solution: three speeds, one decision

REEFPRINT/KHANYA reads the ore at three speeds and turns all three into one decision (Section 3):

1. **the belt** (REEFPRINT), in seconds: a hyperspectral camera above the mill-feed conveyor feeds a model that predicts each parcel’s grinding hardness with an upper bound, and Bond’s law turns the bound into a mill feed-rate proposal;
2. **the laboratory** (KHANYA), within the shift: a microscope model identifies the sulphide minerals and how well they are liberated, which is the brief’s three-phase deliverable;
3. **the decision**, within minutes: a metallurgist or controller approves each proposal before the ore reaches the mill, or the site’s conservative setting applies, and every step is signed and recorded.

It is important to note what is not claimed: performance on South African ore, a measured plant benefit, mineral identification by the belt camera, or identification of chromite and the silicates by the microscope.

1.3 How this report answers the brief and the judging criteria

Table 1 maps the brief’s requirements and the five judging criteria [8] to the evidence in this report.

Table 2: Datasets. All are public; none was supplied by Mintek or a producer.

Dataset	Ore	Size and use	Licence
HIDSAG [9]	Porphyry Cu–Mo, Chile	146 drill-core composites with Bond W_i ; VNIR (39 bands) and SWIR (75 bands). Belt hardness model	CC0
LumenStone S2 [10]	Ni–Cu sulphide, Norilsk	49 micrographs: 37 train, 12 held-out test. Microscope phases	Research use
Bushveld assays [11]	Bushveld chromitites	1,112 intervals. Check on PGE routing	CC BY 4.0

2 How the system works

2.1 Data

No South African belt scans or labelled Bushveld micrographs were available, so every model was trained and tested on public data (Table 2). HIDSAG supplies hyperspectral scans of Chilean porphyry copper–molybdenum drill core with laboratory Bond work-index values [9]. LumenStone S2, a Norilsk nickel–copper sulphide ore [10], shares the mineral assemblage of Bushveld ores but not their abundance: sulphides cover 62.8% of its sections by area, whereas Merensky ore contains up to about 3% base-metal sulphides and UG2 less [5]. Nevertheless, it is the right teacher for the phases, because the base-metal sulphides in UG2 are predominantly pentlandite, chalcopyrite, pyrrhotite and pyrite [5].

2.2 The belt layer: hardness from the infrared

Sensor (design). The design uses a short-wave infrared line scanner such as the Specim SX25 (960–2500 nm, 640 pixels across, up to 162 lines a second) with a visible–near-infrared companion camera [12]. On a 1.2 m belt moving at 2 m/s it records a line every 12.3 mm with 1.88 mm pixels. It therefore averages over parcels of ore; it does not map individual particles.

Model. Each parcel’s spectrum is reduced to features from the infrared absorption regions of clay, mica, chlorite and carbonate minerals (Fig. 3), and a model predicts the Bond work index. Results are reported only for samples the model never saw (out of fold), and a result counted only if it beat simple baselines under pass marks written down before each run.

Bound and gate. The plant setting uses an upper bound that the true hardness should stay below 9 times out of 10 (a one-sided 90% split-conformal bound [13]), not the best guess. A more data-efficient bound [14] was tried first and failed its pass mark, so the exact bound that the plan named as its fallback is used, and the result is provisional. A second check flags parcels unlike anything in training; flagged parcels receive the conservative setting.

2.3 The decision layer

The decision layer turns the hardness bound into a mill setting that a person can approve. Ball mills are usually limited by motor power, so the tonnes a mill can grind depend on

how hard the ore is. Bond’s law gives the specific grinding energy [15]

$$W = 10 W_i \left(\frac{1}{\sqrt{P_{80}}} - \frac{1}{\sqrt{F_{80}}} \right) \text{ kWh/t}, \quad (1)$$

where W_i is the work index and F_{80} and P_{80} are the feed and product sizes (μm) passed by 80% of the material. At a fixed grind target the feed rate the mill can carry is proportional to $1/W_i$, so the proposal is

$$F_{\text{proposed}} = F_{\text{design}} \times \frac{W_{i,\text{design}}}{W_{i,\text{bound}}}. \quad (2)$$

Power, feed size and grind target cancel, so no plant parameter has to be assumed. For example, a design hardness of 16.6 kWh/t and a bound of 16.9 kWh/t give 98.2% of design feed. The proposal must sit inside an operating envelope set by the site’s metallurgist. The demonstration envelope is 85–110% of design feed, with 100% as the conservative setting; feed may be cut by 10% but raised by only 3% per parcel.

A person may approve, modify within the envelope, reject or escalate. If nobody approves before the parcel reaches the mill (an assumed 90 s), or the parcel is flagged, or the data are stale, the conservative setting applies. The system never holds the last value, because the last value may itself be wrong. Every event is written to a tamper-evident, hash-chained record with conventional (Ed25519) and post-quantum (ML-DSA-65 [16]) signatures. The microscope’s advice reaches a simulated regrind setting over OPC UA, the industrial data-exchange standard [17]; the same interface is designed for the feed rate.

2.4 The laboratory layer: minerals under the microscope

The microscope model is a standard image-segmentation network (DeepLabV3 with a ResNet-50 backbone [18, 19]) that labels every pixel of a reflected-light image as resin, pyrrhotite, chalcopyrite, pentlandite or magnetite. The approved checkpoint (SHA-256 de7135a9...) is the only one allowed to send advice to the simulator. A *Quick* scan classifies six 512-pixel fields of a section; a *full* scan covers the whole section.

Connected regions of the mask are split into grains. Chalcopyrite and pentlandite are the valuable (payload) phases, and pyrrhotite is rejected. A grain is *free* if at least half its area is payload and *locked* otherwise; the 50% line is the boundary between low and high middlings in the liberation classes used by Mintek’s characterisation group [2]. A rule-based advisor turns these measures into *grind finer*, *continue* or *verify before acting*, but only if the checkpoint is approved, the image passes a colour check, at least 9 payload-bearing grains were found and mean confidence is at least 0.85. Three of its four thresholds are placeholders for site metallurgists to set.

2.5 How we tested it

Belt models were scored out of fold against the strongest simple baseline (the fold mean, the fold median and a colour-only model), with pass marks committed to the repository

before each run. The microscope was scored on the publisher’s 12 held-out S2 sections, never used for training or model selection. Its accuracy is the intersection over union (IoU: the overlap between predicted and expert-labelled areas, from 0 to 1), pooled over 103.8 million pixels and averaged per section, with bootstrap confidence intervals over sections. Decision-level agreement runs the advisor on the model’s mask and on the expert’s mask; a disagreement is *unsafe* if the model gives a confident instruction that the expert mask does not support, for example *continue* where the expert mask says *grind finer*. A request to verify is safe.

3 System architecture

Figure 2 illustrates the system, highlighting that both layers pass the same safety gates and the same human decision. The belt monitor (REEFPRI Live) runs the hardness model, the decision view and the signed record. The microscope workbench (KHANYA) is a web application (React) on a Python service (FastAPI and PyTorch) that runs on an ordinary laptop processor. In local mode it accepts connections only from the same machine and makes no external requests, so a network outage at the plant does not stop the advice. No language model computes a measured value or a control value.

Speeds were measured on a laptop processor: 92 ms per belt parcel (median, four cores); 16.7–21.2 s per six-field microscope scan on mains power (5 runs) and 30.8 s on battery; 162 s for a full section. The 90 s approval window is an assumed belt-to-mill transit time. Three environments exist, and each claim names its environment: a static public demonstration of the belt monitor on precomputed data with an unsigned record; a secure server (role-based access, password hashing, request forgery protection, rate limits and the signed record) that passed 17 of 17 security tests locally and is not deployed; and the microscope workbench, which ran locally. The workbench has 180 unit-test functions and 7 browser tests, with four continuous-integration checks passing, and the belt software has 328 unit-test functions; static security analysis reported no issues after a dependency upgrade.

4 Results

4.1 Belt: predicting grinding hardness

Of the five HIDSAG targets tested (copper and molybdenum recovery, lime consumption, pH and Bond work index), only the work index beat its strongest baseline under every pass mark. Figure 4 illustrates the out-of-fold results, highlighting that the infrared carries most of the information: the hyperspectral model explains 48% of the variance in work index, while a model given only colour explains 21% and the average of the training data explains nothing. Its 80% intervals covered 79.5% of samples, as intended.

Figure 3 illustrates one parcel as the model sees it, highlighting the absorption features the model reads. For parcel GMET-0004 the predicted W_i is 15.9 kWh/t and the 90% bound is 16.9 kWh/t, against a design hardness of 16.6 kWh/t (the training data’s 90th percentile), so the proposal is 98.2% of design feed. The laboratory value was 12.1 kWh/t: the

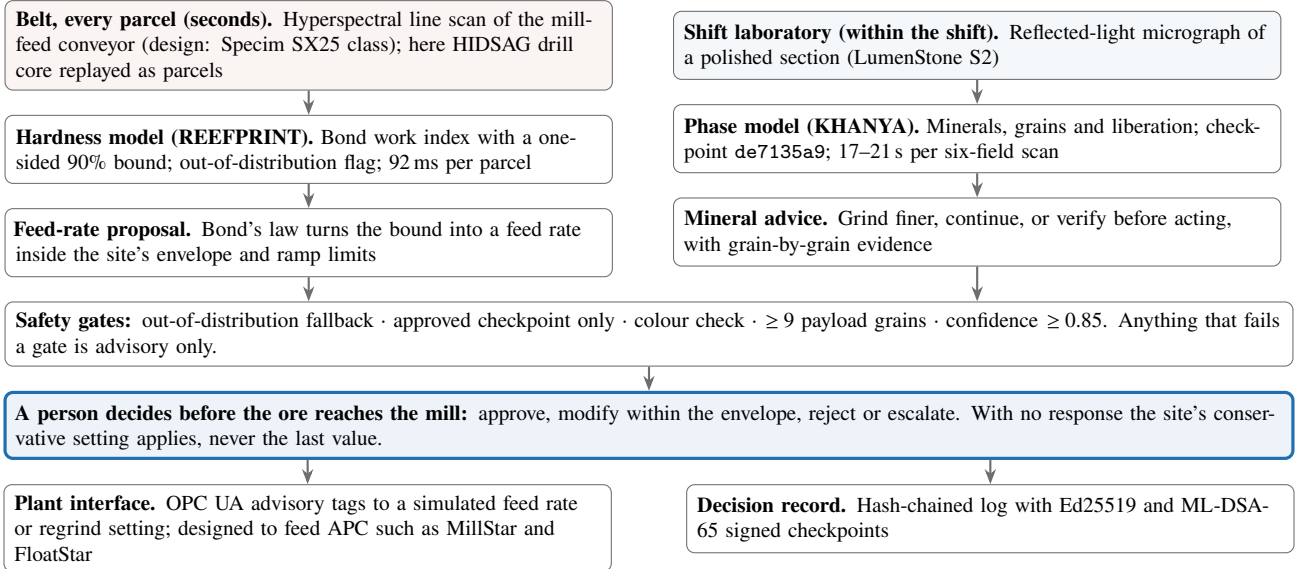


Figure 2: System architecture: three speeds, one decision. The belt layer (left) is the site system; the microscope layer (right) is the laboratory truth layer and delivers the brief’s three phases. Both pass the same gates and the same human decision. Everything runs offline on one laptop processor; the plant is a simulator.

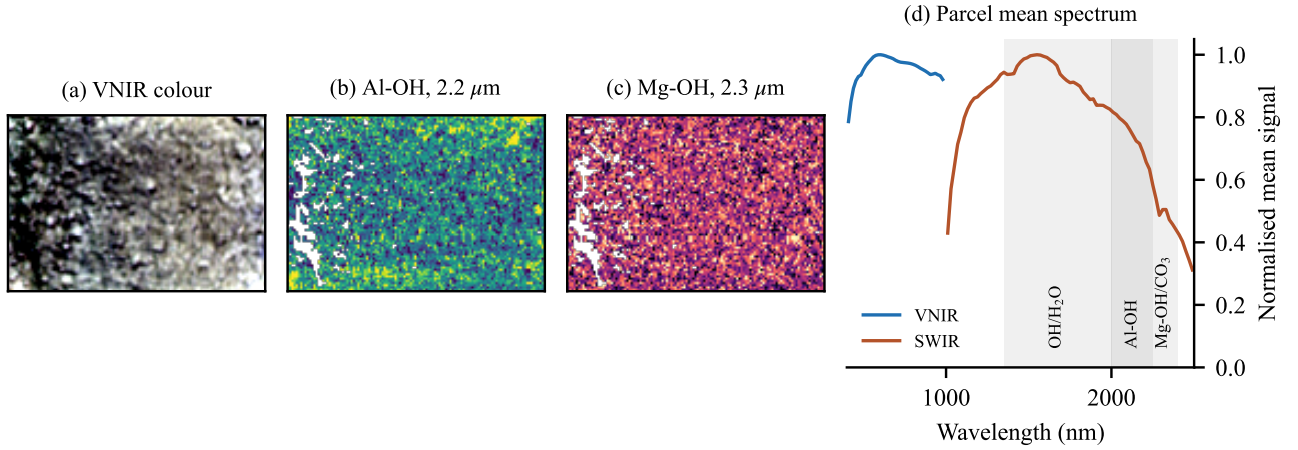


Figure 3: One belt parcel as the hardness model sees it: HIDSAG composite GMET-0004 (Chilean porphyry Cu–Mo), 80×117 pixels. (a) Colour image from the visible bands. (b, c) Relative absorption depth at the Al–OH (white mica, clays) and Mg–OH (chlorite, biotite, carbonate) features; detector stripes removed for display only. (d) Mean spectrum, with the absorption regions the model uses. The model predicts W_i 15.9 kWh/t with a 90% upper bound of 16.9 kWh/t (laboratory value 12.1 kWh/t). The maps show absorption features, not a mineral identification.

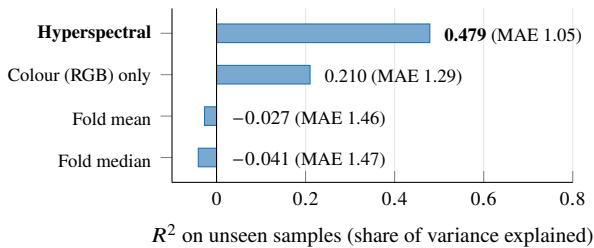


Figure 4: Bond work index on 146 HIDSAG composites, out of fold; mean absolute error (MAE) in kWh/t. Mean measured W_i is 14.62 kWh/t.

bound was conservative here, which is the direction a safety bound should err.

4.2 Belt: from hardness to a plant decision

Figure 5 shows the decision for the same parcel as the operator sees it. Figure 6 illustrates four feed-rate policies replayed over all 146 composites, highlighting the trade-off between throughput and overloads. Setting the feed from the bound gave a simulated throughput gain of +1.9% (95% CI +0.9 to +2.9%), and overloaded parcels fell from 8.9% to 6.8%; the one-sided 95% upper bound on the change (+1.4 points) is inside the +3-point margin set before the test. Feeding from the best guess instead would add 13.6% throughput but overload 45.2% of parcels, which is why the policy uses a bound. Perfect knowledge would add 14.0%, so the bound captures about one-seventh of what is available, and better hardness models have room to add value.

The result is provisional for three reasons: the exact bound was adopted after the more efficient one failed, HIDSAG has no time order (so transit time, stockpile mixing and ramp

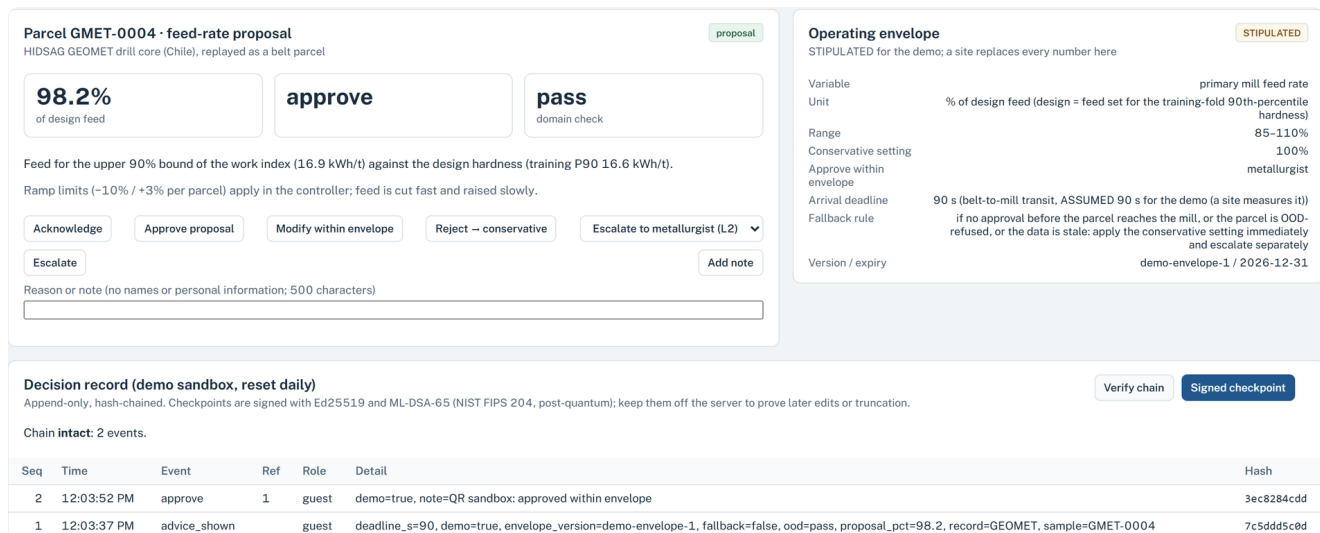


Figure 5: The Decisions view of REEFPRINT Live for parcel GMET-0004: a feed-rate proposal of 98.2% of design inside a demonstration envelope, the operator’s actions, and the hash-chained decision record with a guest approval (demonstration sandbox). Screenshot of the application.

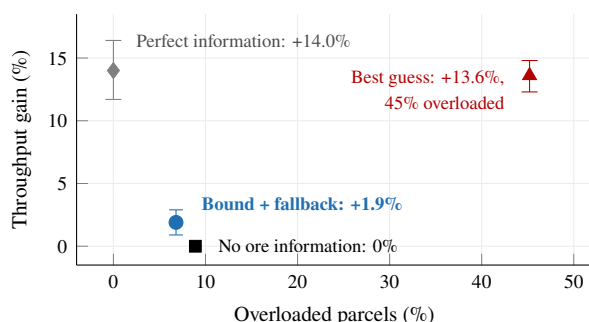


Figure 6: Feed-rate policies replayed over 146 composites (simulated; not PGM ore and not a time series), with 95% confidence intervals. Up is more throughput; left is fewer overloads. The bound adds throughput without adding overloads.

limits are not simulated), and the ore is not Bushveld ore. A robustness test, also defined before it was run, found that a new capture of the same ore keeps its error (mean absolute error $\times 1.00$, 95% CI 0.99–1.01). Half the field of view ($\times 1.01$), added noise ($\times 1.00$) and $\pm 15\%$ illumination ($\times 1.01$) were also tolerated. However, a wavelength drift raised the error by a factor of 1.25 and was mostly not flagged, so a wavelength-calibration check was added to the hardware design.

4.3 Is it physically possible?

Every claim was tested against its physics, and the failures stay in the report (Table 3). The most useful negative check concerns load-shedding: re-ordering ore during electricity curtailment cannot create tonnes, because energy per tonne is conserved over a cycle, so that claim was rejected.

4.4 Laboratory: phase identification

The microscope meets the brief’s three-phase requirement. Figure 7 illustrates its accuracy by class, highlighting IoU values of 0.87 for pyrrhotite, 0.58 for chalcopyrite and 0.55 for pentlandite. Magnetite, a fourth class the brief does not require, is never predicted: 78.3% of it is labelled as the

Table 3: Physical-possibility checks. Modelled values are labelled; none is a plant measurement.

Claim	Result	Status
The bound gains tonnes (Bond’s law)	+1.9% [+0.9, +2.9], simulated	Provisional
Without losing flotation recovery	Modelled cost ≤ 0.25 points on the steep part of the curve; none on the plateau	Check residence first
More tonnes under load-shedding by re-ordering ore	−3.8% simulated; even perfect knowledge gives −3.0%	Rejected
Works on an unseen capture	Error $\times 1.00$ [0.99, 1.01]	Holds
Fast enough for the belt	92 ms per parcel, laptop processor	Measured
Camera resolves the belt	12.3 mm lines, 1.88 mm pixels: parcels, not particles	Computed
Belt camera identifies minerals	No test passed; the microscope carries the phases	Not shown
Radiation	Optical only: no radioactive source and no radiation sensing	—

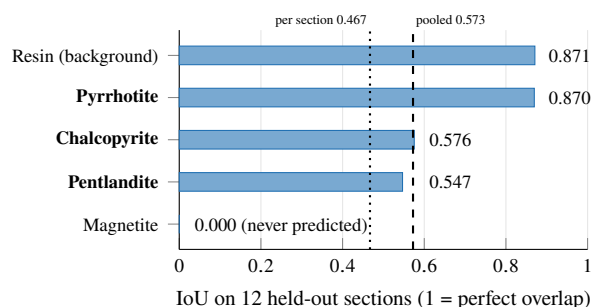


Figure 7: Microscope accuracy by class for checkpoint de7135a9, full-section inference. The three phases the brief requires are in bold. Section means range from 0.340 to 0.728.

dark mounting resin. The pooled mean IoU is 0.5725 (95% CI 0.494–0.624); averaged per section it is 0.4671 (0.412–0.534), and 10 of the 12 sections fall below the pooled figure. The model clearly beats trivial baselines (majority class 0.117; colour-only 0.395). A second training run of the same recipe scored 0.454 because the patch sampler was not

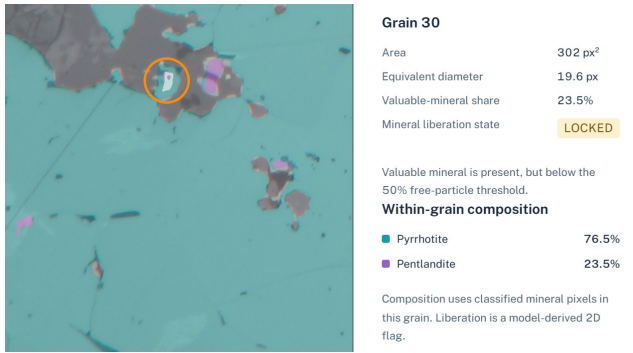


Figure 8: The grain inspector on held-out section test_11 (Quick scan, checkpoint de7135a9): one analysed field with the phase overlay (left; teal pyrrhotite, violet pentlandite) and the selected grain’s evidence (right). Grain 30 (circled) is 76.5% pyrrhotite and 23.5% pentlandite, so it is locked; grain 31, the violet grain to its right, is pure pentlandite and free. Cropped from a screenshot of the application.

seeded, so these figures describe the approved checkpoint, not the method. A retrained candidate scores 0.632 overall but loses accuracy on pyrrhotite and pentlandite, the phases that matter most for PGM deportment, so it is not deployed.

4.5 Laboratory: decisions and grain evidence

Running the advisor on the model’s masks and on the expert masks, the shipped pipeline disagreed on 8 of 12 sections, and every disagreement was a request to verify: none was unsafe (the unrefined pipeline made 2 unsafe errors). Zero in 12 is an observation, not a rate; the one-sided 95% upper bound on the unsafe rate is 22.1%. A six-field scan matched the whole-section advice on 9 of 12 sections, against 0 of 12 for a single centre field. However, a realistic exposure shift (−35 levels on each colour channel) changed the advice on 8 of 12 sections, so illumination control is a pilot requirement.

Figure 8 illustrates the grain inspector, highlighting how a mineralogist can audit the advice grain by grain. In Quick mode each field is classified without context beyond its edge, and on test_11, 19 of 24 grains come within 6 pixels of a field edge. Thus, Quick-mode grain statistics are indicative, and the full scan is the reference.

5 Validation against literature and limitations

5.1 Validation against literature

The results align with published work and with what producers already report. Notole et al. [20] validated hyperspectral imaging and portable XRF against reference methods on Merensky Reef drill core scanned at the Council for Geoscience in Pretoria, so the sensing layer has been shown to work on Bushveld material, although not yet for hardness. Our 50% free/locked line sits on the boundary between low and high middlings in Mintek’s liberation classes [2], and the three sulphides the microscope identifies are the main base-metal sulphides of UG2 ore [5]. On the plant side, the effects we test for are smaller than those already reported. Mintek’s AI optimisation raised a lead–zinc concentrator’s throughput by 0.5%, worth over US\$2 million (about R33 million)

a year [21], and Valterra’s recovery gains of 1.2–1.7 percentage points [6] are many times our break-even (Section 6.3).

Four ideas were tested and withdrawn, and they mark where the method stops. The belt spectra did not predict QEMSCAN mineralogy once size fraction and process line were accounted for (no skill on any of 33 minerals). Belt chemistry routed Bushveld PGE ore worse than the mine plan’s seam label (balanced accuracy 0.79 against 0.82). The belt did not predict recovery, lime or pH better than a baseline. Plant tags never forecast concentrate silica an hour ahead better than the last laboratory assay.

5.2 Limitations

The following limitations apply: (a) no South African ore has been evaluated, because the belt data are Chilean copper core and the microscope data are Norilsk ore; (b) the belt result is simulated on 146 composites without time order, so it is provisional until a pilot measures it; (c) the belt camera predicts hardness and does not identify minerals, while the microscope has not been trained on chromite or the silicates and never detects magnetite; (d) the 12 held-out sections were reused during development, and the training recipe is not exactly reproducible; (e) the microscope’s advice is sensitive to illumination, three advisor thresholds are unsourced, and liberation is measured in 2D, which overstates it; and (f) latency is measured on one laptop with small samples, and the plant is a simulator. Compared with the abstract Mintek accepted, three things changed: the five Bushveld phases became three sulphides on an analogue ore, because no labelled Bushveld images existed; the QEMSCAN “teacher” became the pilot’s truth-laboratory stream; and the fallback was reversed from holding the last setpoint to applying the conservative setting.

6 Business model and financial case

6.1 Value proposition and business model

Our value proposition is simple: the plant gets a reading of its ore before the mill, with a stated bound, and pays only for value that an independent party has measured. The buyer is the concentrator manager or general manager for metallurgy at a South African PGM producer, the person who owns recovery, concentrate quality and mill throughput. Producers such as Valterra, Implats, Sibanye-Stillwater, Northam and Tharisa are named as market segments, not customers. Figure 9 illustrates how money and value flow between the partners, highlighting that the producer’s fee is tied to measured value. The Technology Innovation Agency (TIA), the national agency that funds technology development [22], is the proposed bridge funder.

The revenue model has four streams: a paid laboratory pilot; a monthly licence for the laboratory software; recovery of the belt pilot’s cost; and, after a successful pilot, at most 20% of the measured net value per concentrator per year. The model is justified for four reasons, each drawn from Mintek’s own published work [21]:

1. **It creates more truth-laboratory work, not less.** The models need QEMSCAN composites and Bond tests to retrain and audit them, and Mintek runs QEMSCAN [2]. Mintek already characterises drill core for chromite and

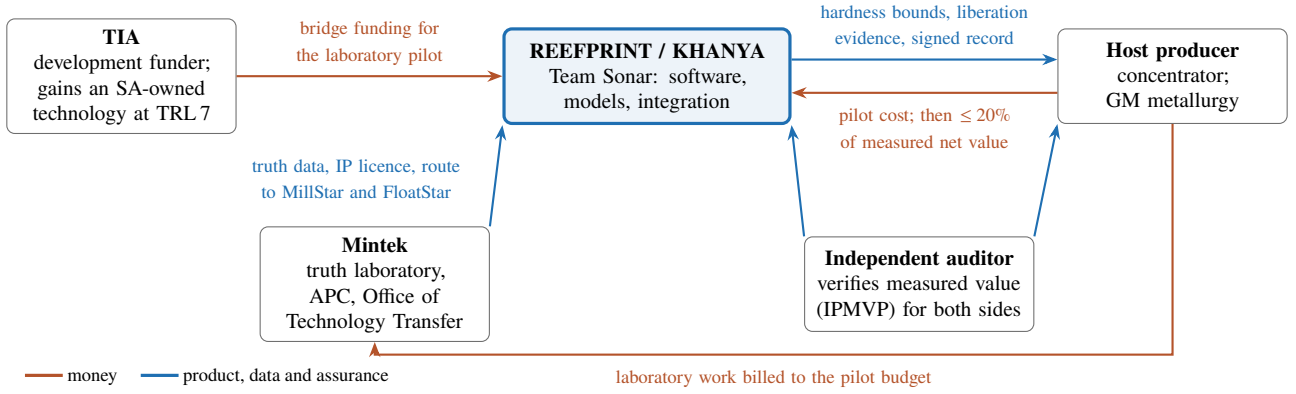


Figure 9: The business model. The producer pays for a pilot and then a capped share of value that an independent party has measured, so no measured value means no fee. Mintek earns from the laboratory work the models need and licenses the IP through its Office of Technology Transfer (MOTT); TIA’s bridge funding de-risks the laboratory pilot.

Table 4: What plants use today, and where this system differs.

Option	What it gives	Where this differs
Cross-belt neutron analysers	Bulk elements, minute by minute	Hardness and minerals, not elements; isotope sources need a SAHPRA licence [24], this needs none
Slurry analysers	Grade and Cr_2O_3 in slurry, after milling	Reads the ore before the mill; their output is an input
Automated mineralogy (QEMSCAN)	Phases and liberation, in days	Predicts between truth results; its composites retrain the models
APC (e.g. MillStar, FloatStar)	Stable, optimised circuits from process data	A partner: an ore-side input inside APC limits

iron-ore washability and applies ore characterisation to flotation reagent selection for PGMs and base-metal sulphides.

2. **It is a feed-forward input to Mintek’s process control.** Mintek’s AI-driven optimisation raised a concentrator’s throughput by 0.5%, worth over US\$2 million a year; a hardness bound that arrives before the ore is the kind of input such optimisation can use.
3. **Mintek has already proven value-linked pricing.** Its benefit-sharing model with Hindustan Zinc in India links revenue to system performance and has generated over R35 million since 2021.
4. **It has a route beyond South Africa.** Mintek’s products and services are used in over 60 countries, its instruments group operates in 42, and it works through distributors including Molycop and Mercantil. Its footprint includes Zimbabwe, Zambia, the DRC and Chile, where the belt model was developed.

The timing also favours a pilot now: South African PGM output fell 9% in 2025 [4], while Valterra’s realised basket price rose from R32,611 per ounce in 2025 to R45,993 in the first half of 2026 [6, 23], so each ounce recovered is worth 41% more.

Table 4 compares the options a plant can buy today. The system does not replace any of them; it adds an ore-side input before the mill and uses their results as its truth.

6.2 Pricing, and what the money pays for

Table 5 sets out the price of each stage and, more importantly, what it pays for beyond equipment and software: sampling, laboratory truth, maintenance, training, security review and independent verification. The funding split follows the partners’ roles. TIA bridge-funds the laboratory pilot; the host producer pays the belt pilot’s installed and running costs; and Mintek’s laboratory work is billed to the pilot budget.

6.3 Financial methodology

No plant benefit has been measured, so the case is a worked example with stated assumptions: (a) annual output is that of Northam’s Zondereinde mine, 333,050 equivalent refined 4E ounces in F2026 [26]; (b) metal prices are Valterra’s realised basket prices of R32,611 per ounce (2025) and R45,993 (first half of 2026) [6, 23], used as proxies; (c) value is revenue before incremental processing and refining costs, so it overstates profit; (d) costs are the ranges in Table 5; and (e) the pilot equipment has a 5-year life at a 10% discount rate.

The annual value of an uplift u in recovered metal is

$$V = u \times Q \times P, \quad (3)$$

where Q is annual output in ounces and P the price per ounce. For $u = 1\%$,

$$V = 0.01 \times 333,050 \times (\text{R}32,611 \text{ to } \text{R}45,993), \quad (4)$$

which is equivalent to R108.6–153.2 million a year. The installed cost is spread over the equipment’s life with the capital recovery factor

$$\text{CRF} = \frac{r(1+r)^n}{(1+r)^n - 1} = 0.2638 \quad (r = 10\%, n = 5), \quad (5)$$

so the annualised cost is

$$C = \text{CRF} \times C_{\text{installed}} + C_{\text{running}}, \quad (6)$$

which is equivalent to R3.8 million a year (low), R11.4 million (middle) and R22.8 million (high), or R25.4 million for the high case with contingency. The break-even uplift is then

$$u_{\text{BE}} = \frac{C}{Q \times P}, \quad (7)$$

Table 5: Pricing by stage. Prices are the team’s hypotheses (H) or planning assumptions (A) until vendors and laboratories quote. Ranges first set in US\$ are converted at R16.4 per US\$, the January–September 2026 average [25].

Stage	Price	What it pays for	Who pays	Basis
1. Laboratory pilot (KHANYA), 3 months	About R210,000	Team time to retrain the microscope on QEMSCAN-labelled sections of the host’s ore; a validation report against QEMSCAN; workbench installation; training of laboratory staff	TIA bridge funding	H
2. Laboratory licence	About R12,000 per site per month (R144,000 a year), plus a deployment fee	Software updates, model audits, support and security patches	Host laboratory	H
3. Belt pilot, one concentrator, 9 months	R8.2–49.2 million installed; R1.6–9.8 million a year to run	<i>Installed:</i> infrared and visible cameras, enclosure and window, lighting, calibration references, belt-speed encoder, edge computer with UPS, structural and electrical work, OPC UA and historian integration, an OT security assessment, commissioning. <i>Running:</i> sampling, assays, Bond tests, QEMSCAN composites, maintenance and spares, support, training, independent measurement and verification	Host producer; Mintek’s laboratory work billed to the pilot	A
4. Contingency	20% of installed cost (R1.6–9.8 million)	Released only against quotes at gate G1	Held by the funder of each stage	A
5. Scale-up	At most 20% of measured net value per concentrator per year	Models, audits, support and upgrades; zero if no value is measured	Producer	H

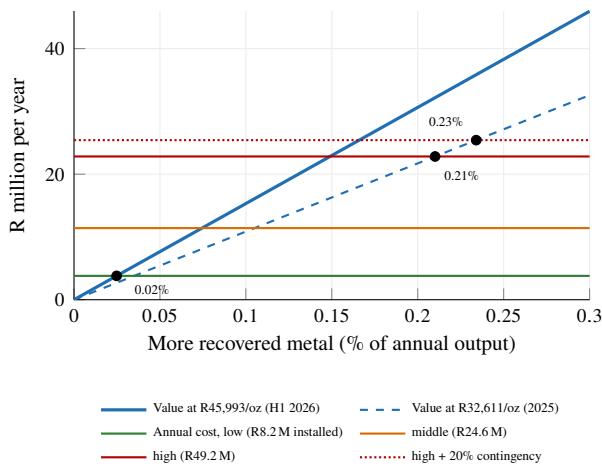


Figure 10: Break-even for one belt at Zondereinde’s output: the annual value of extra recovered metal (blue) against the annualised pilot cost (horizontal lines). Dots mark the best case, the worst case and the worst case with contingency.

which gives 0.02% in the best case (low cost, 2026 price) and 0.21% in the worst (high cost, 2025 price), or 0.23% with contingency. Figure 10 illustrates the break-even, highlighting that every crossing lies below a quarter of a per cent.

The decision is therefore insensitive to where a quote lands in the cost range. A 10% weaker rand moves the worst case from 0.21% to 0.23%, and a basket price 30% below the 2025 level moves it to 0.30%. For scale, if the pilot measured 0.25% more recovered metal, the value would be R27–38 million a year before incremental costs, and at most a fifth of the net figure would be due as a fee.

6.4 Risks, contingency and fallbacks

Figure 11 and Table 6 set out the main risks of using the product, as the team assesses them. Each has a mitigation that acts before it happens and a fallback that limits the damage if it does. Three fallbacks run through all of them: the plant always has its conservative setting and its existing workflow; each pilot phase ends at a gate where spending can stop; and

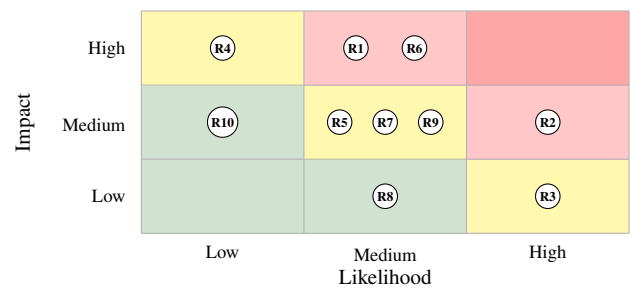


Figure 11: Risk matrix (team’s assessment before mitigation). Risks R1–R10 are described in Table 6.

the fee is zero if no value is measured.

7 Implementation and support

7.1 Support needed

The pilot depends on people more than on hardware. Table 7 lists the roles it needs, why, when and how much of their time, as planning assumptions to agree with the host site.

7.2 Maintenance

The design keeps routine upkeep simple and makes most checks automatic (Table 8). It is important to note that the system refuses rather than guesses when a check fails, so poor maintenance shows up as more conservative settings, not as wrong advice.

7.3 Compliance, committees and independent checks

Table 9 lists the laws, policies and standards that apply, what each asks of the project and how the design complies. The most important simplification is that the camera carries no radioactive source, so no radiation licence is needed [24].

Committees. Before anything is installed, four site bodies sign off: the health and safety committee (MHSA), the

Table 6: Risk register: mitigation before the event, and the contingency or fallback if it happens.

#	Risk	Mitigation (before)	Contingency or fallback (if it happens)
R1	The hardness model does not transfer to Bushveld ore	Laboratory pilot on the host's ore first; shadow phase compares bounds with site Bond tests (G2)	Retrain on site data, or stop at month 5; spending is limited to that point
R2	Belt conditions: dust, moisture, vibration, lamp ageing, wavelength drift	Purge air, calibration references and a calibration gate at commissioning (G1)	Flagged parcels get the conservative setting; the plant runs as it does today
R3	Power outages and load-shedding	Runs offline on an edge computer with a UPS	After a restart, the conservative setting holds until checks pass
R4	Cyber attack or an unsafe write to the plant	Read-only advisory tags; no write path to safety systems; IEC 62443 zones; signed records; offline backups	Disconnect; the plant continues on its existing control system
R5	Operators ignore or over-trust the advice	Training before the advisory phase; named approval roles; a fixed approval window	Escalation; the conservative setting applies when nobody responds
R6	No measurable gain	On/off operating blocks with confidence intervals; independent measurement and verification	Stop at G3; no fee is due
R7	Cost overrun or a weaker rand	Quotes before G1; 20% contingency; funding in tranches at gates	Defer the belt pilot and continue the laboratory pilot
R8	A fall in PGM prices	Break-even stays at or below 0.30% even at a 30% lower basket price	The value share falls with value automatically
R9	Funding delay	The laboratory pilot is small (about R210,000) and can start first	Laboratory pilot only; the belt pilot starts when funded
R10	Privacy or labour objections to the decision record	POPIA purpose, retention and non-disciplinary use agreed with the information officer and workforce representatives	Record approval roles instead of names

Table 7: Support needed from the host site, Mintek and the funders. Time commitments are planning assumptions.

Role	Why the pilot needs them	When (month)	Commitment
Plant manager or GM metallurgy	Sponsor; owns the budget, approves the envelope, decides at each gate	0–9	Gate meetings
Process metallurgist	Sets the envelope and the conservative setting; approves proposals; reviews the record	0–9	0.3 full-time
Mineralogist (Mintek)	QEMSCAN truth; labels sections for retraining	0–9	0.2 full-time
Analytical chemist	Assays and Cr ₂ O ₃ ; quality control with blanks, duplicates and reference materials	2–9	Per sampling plan
Process control engineer	OPC UA tags, historian access, later coupling to APC	1–2, then 9+	As needed
Instrumentation engineer and technician	Installation, enclosure, references, lamps	1–9	0.3 full-time
OT and IT security officer	Network zone, read-only interface, backups	0–2	0.1 full-time
Health and safety representatives	Risk assessment before installation (MHSA)	0–1	Consultation
Information officer and workforce representatives	POPIA and the decision-record policy	0–1	As needed
Mintek MOTT; TIA portfolio manager	IP agreement; funding tranches at gates	0, gates	As needed
Team Sonar	Models, software, integration, analysis and reporting	0–9	Three full-time roles

management-of-change review, the IT/OT change board and the capital investment committee. On the funding side, the pilot passes TIA's investment process and Mintek's MOTT.

Independent fact-checkers to outsource. Six checks should be done by parties other than the team: (a) a SANAS-accredited laboratory for Bond tests and assays; (b) Mintek's QEMSCAN for mineralogy; (c) an independent measurement-and-verification auditor for the value on which fees are charged; (d) an IEC 62443 assessor for operational-technology security; (e) an IP attorney through MOTT; and (f) a university reviewer for the model audit.

7.4 Employee training

The MHSA requires training before significant changes to plant or equipment, and a record of it; mines also submit workplace skills plans and training reports to the Mining Qualifications Authority (MQA) [27, 36]. The modules in

Table 10 are therefore aligned with the site's workplace skills plan, so that they count toward its MQA reporting and its Mining Charter skills spend.

8 Roadmap after funding approval

Figure 12 illustrates the plan from the day funding is approved, highlighting that each phase ends at a gate the partners agree in advance. If a gate fails, the pilot stops, and the spending to that point is the maximum exposure. Table 11 lists the milestones and the evidence that each must show.

After G3 the system is coupled to the pilot site's process control within its limits (months 9–12), and the product is hardened in parallel. A second site on another reef tests transfer by month 18. The South African rollout then proceeds producer by producer, and from month 24 Mintek's distribution network offers a route to Zimbabwe's Great Dyke and the Copperbelt.

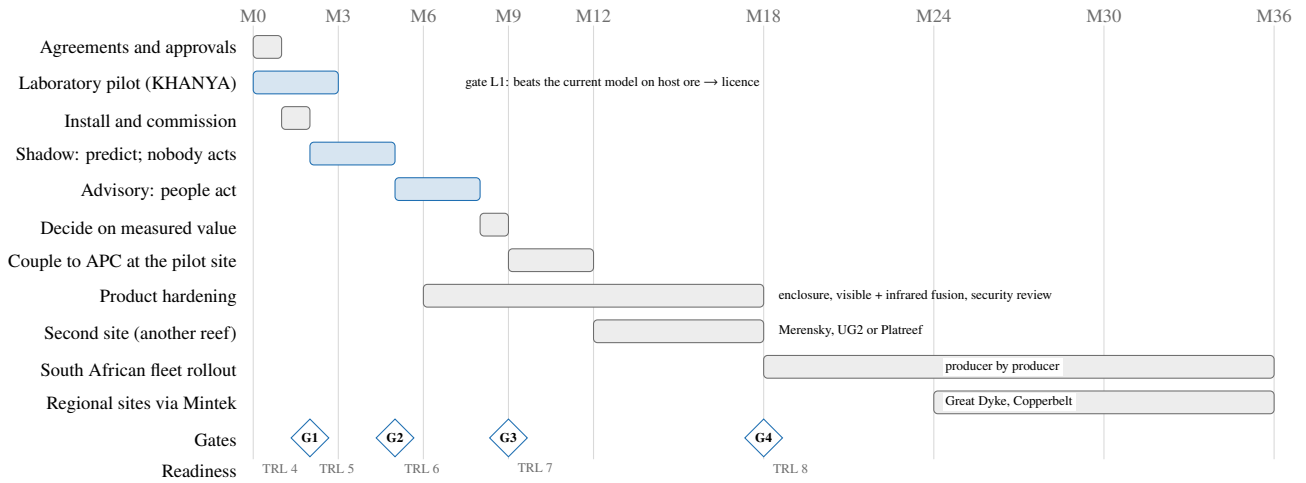


Figure 12: Roadmap from funding approval (M0) to month 36. Shaded bars are the nine-month pilot’s core phases; diamonds are go/no-go gates (Table 11). TRL levels are the team’s self-assessment. Months beyond M9 depend on G3.

Table 8: Proposed maintenance plan; the site sets the final intervals.

Task	Interval	Owner
White and wavelength reference checks; refuse if a feature moves by more than half a band	Every reference cycle (automatic)	System
Window cleaning and purge-air check	Each shift, or on a fouling alarm	Instrument technician
Lamp replacement	Lamp-life schedule	Instrument technician
Enclosure temperature and dew point	Continuous alarm	System; electrical
Microscope colour check; illumination calibration	Every image; daily	Laboratory technician
Model audit against Bond tests and QEMSCAN	Monthly in the pilot	Metallurgist, mineralogist
Retraining	Only through a validation gate	Team Sonar
Security patches; backup restore test	Monthly; quarterly	Team Sonar, IT/OT

9 Conclusions and recommendation

A hyperspectral camera on the belt can predict grinding hardness well enough to steer a mill: on 146 public drill-core composites the model explained 48% of the variance in Bond work index, more than twice what colour alone explains, and a feed-rate policy built on its upper bound gave a provisional simulated throughput gain of 1.9% without more overloads. A microscope model identified the three main base-metal sulphides of UG2 ore on held-out sections, with magnetite as a stated failure, and gave no unsafe advice on 12 sections. Both feed one decision that a person approves and a signed record keeps. These results hold for public analogue ores and a simulated plant.

The financial case does not depend on large effects. At Zondereinde’s output a one-belt pilot breaks even at 0.02–0.21% more recovered metal, a small fraction of the recovery moves producers already report, and the producer pays a share of value only once an independent party has measured it. We therefore recommend that TIA, Mintek and one PGM producer approve a nine-month pilot: one belt and one truth laboratory, with go/no-go gates at months 2, 5 and 9. Whether the results transfer to South African ore is exactly

the question that pilot is designed to answer.

CRedit authorship contribution statement

Lethabo Hoaeane: Conceptualisation, domain lead, belt hyperspectral modelling, belt monitor and decision application, economics, presentation. **Sibusiso Khumalo:** Microscope segmentation model, accuracy and decision-level evaluation, advisor and safety gates, OPC UA simulation, workbench review and verification, technical report.

Declaration of competing interest

The authors declare no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Use of AI tools

Code and drafting were assisted by AI coding assistants. Every reported number was computed by code in the project repository and checked by the authors.

Data and code availability

HIDSAG is available under CC0 (doi:10.6084/m9.figshare.c.5983921) [9]; LumenStone under research-use terms [10]; the Bushveld assays under CC BY 4.0 (doi:10.17632/dc8jcnbcvk.1) [11]. Code, checkpoint hashes and every results file behind the numbers in this report: <https://github.com/Sibusiso-K/KHANYA> (belt: branch `codex/pwa-phase-roadmap`; microscope: branch `codex/launch-live-demo`) **[VERIFY: the repository is private on 5 Oct 2026: make it public, or change this line, before submitting]**. A static demonstration of the belt monitor runs at <https://lethabomh14-reefprint.static.hf.space>.

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Table 9: Laws, policies and standards that apply to the pilot, and how the design complies.

Law, policy or standard	What it asks of this project	How the design complies
Mine Health and Safety Act 29 of 1996 [27]	Assess risks of new equipment, consulting the health and safety committee (s. 11, s. 34); train employees before significant changes to plant or equipment (s. 10)	Installed under the site's risk assessment and lockout; advisory only; operators trained before the advisory phase
Protection of Personal Information Act 4 of 2013 [28]	Lawful, limited processing of personal information, such as approvers' names in the decision record	Purpose, access, retention and non-disciplinary use agreed with the information officer before the shadow phase
Mining Charter 2018, clause 2.3.1 [29]	Invest 5% of the leviable amount in essential skills development, including R&D in processing and technology efficiency	Pilot training and R&D may count toward it; clause 2.3 was not among the clauses set aside by the High Court in 2021 [30]
IPR from Publicly Financed R&D Act 51 of 2008 [31]	IP from publicly financed R&D is identified, protected and commercialised for South Africa (NIPMO)	IP managed by Mintek's Office of Technology Transfer (MOTT) under the hackathon agreement, with invention credits [8]
Public Finance Management Act 1 of 1999 [32]	Mintek (Schedule 3B) and TIA (Schedule 3A) spend public money under PFMA controls	Funding released in tranches tied to gates G1–G3, with auditable deliverables
Hazardous Substances Act 15 of 1973 (SAHPRA Radiation Control) [24]	Licences for radioactive sources (Group IV)	Not applicable: optical sensing only, unlike isotope-source analysers
IEC 62443 [33]; OPC UA [17]	Zones and conduits for industrial cyber security; secure industrial data exchange	Read-only advisory tags; no write path to safety systems; IT/OT sign-off before installation
ISO/IEC 17025, accredited by SANAS [34]	Demonstrated competence of testing laboratories	Bond tests and assays used as truth come from SANAS-accredited laboratories
IPMVP (EVO) [35]	A recognised protocol for measuring and verifying savings	Fees invoiced only on value measured under an agreed IPMVP-style plan by an independent party

Table 10: Proposed training plan.

Who	What they learn	Format; when
Control-room operators, shift supervisors	Reading a proposal; approve, modify, reject or escalate; what the fallback does	Half a day plus supervised shifts; before month 5
Metallurgists	Setting the envelope; reading bounds and liberation; reviewing the record	One day; month 1
Laboratory mineralogists and technicians	The workbench, colour calibration, grain audit, comparison with QEMSCAN	One day; months 0–3
Instrument and maintenance technicians	Camera, enclosure, references, lamps, lockout	One day with the vendor; at commissioning
IT and OT staff	Network zones, backups, patching, incident response	Half a day; before installation
Managers and funders	Reading the measured-value report and the gates	One hour; at each gate

Table 11: Milestones and go/no-go tests after funding approval.

Gate	Month	Evidence to pass	If it fails
–	1	Agreements signed (data, POPIA, OT access, IP); envelope approved	Do not install
L1	3	Microscope retrained on host ore beats the current model; agreement with QEMSCAN reported	Stay at research level
G1	2	References and calibration pass; no write path to any safety system	Fix before shadow
G2	5	Bound coverage holds against site Bond tests; out-of-distribution rate acceptable	Retrain or stop
G3	9	Measured gain above break-even, with a confidence interval, verified independently	Stop; no fee
G4	18	A second site passes G2 within budget	Stay single-site

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